INCH-POUND

MIL-I-25907C(USAF)
30 JUNE 1994
SUPERSEDING
MIL-I-25907B(USAF)
15 September 1979

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, EFU-4/A, FUEL TOTALIZING, 500-15,000 PPH

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a single remote indicating fuel flow indicator, designated EFU-4/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-1762	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-I-7566	Indicator System, Fuel Flowmeter, General Specification For
MIL-P-8422	Power Supply 28V DC Input, 15V 4 Cycle, 3 Phase Output, Type MC-1
MIL-T-25905	Transmitter, Rate Of Flow, TRU-7/A, Fuel 0-15,000 PPH
MIL-I-25908	Indicator, Rate Of Flow EFU-5/A, Fuel, Totalizing, 0-120,000 PPH

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.1.

3.2 General specification. The requirements specified in MIL-I-7566 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirement of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.3 Design and construction.

3.3.1 Individual rate indication. The indicator shall operate on the second harmonic system in which the moving element consists of non energized permanent two-pole magnet(s) and the toroidally wound stator is energized with 115V, 400-cycle single-phase ac. In addition to the excitation voltages, even harmonics may appear across the winding dependent upon the position of the moving element.

3.3.2 Summation output signal. The indicator shall contain a solid state conversion circuit with no moving parts, to convert the electrical input signal to the indicator into 400 Hz voltage of IV per 1000 pph for use by a remote EFU-5/A.

3.3.2.1 Failure. Failure of one or more individual rate indicators shall not affect totalizing of the system, and failure of the totalizing indicator shall not affect normal individual rate indication.

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3.3.3 Calibration. Each indicator dial shall be graduated in accordance with figure 1. The calibration of fuel flow versus angular degrees shall be linear in each range within ± 0° 15' of arc, except at the changeover point. The 14,000 pph graduation shall be located at the nine o'clock position.

3.3.3.1 Rate of flow. Unless otherwise specified, the indicator shall be designed to indicate, at a remote station, the rate of flow in pounds per hour of aviation fuel when used in conjunction with a fuel flow transmitter in accordance with MIL-T-25905, a summation indicator in accordance with MIL-I-25908, and a power supply in accordance with MIL-P-8422.

3.4 Pointer. The pointer shall conform to MS33585-8.

3.5 Markings.

3.5.1 Fluorescent-luminescent. The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

MARKINGS	HEIGHT OR LENGTH INCH ± 0.016	WIDTH OF LINE OR GRADUATION INCH ± 0.005
Scale Numerals 1, 2, and 3	0.156	0.031
Scale Numerals 6, 9, 12, 15, and 0.5	0.094	0.020
Numbered 1,000-pound graduations 1 through 3	0.156	0.031
Numbered 1,000-pound graduations 6 through 15	0.125	0.020
Intermediate 1,000-pound graduations	0.094	0.020
500-pound graduations 0.5 through 3	0.125	0.031
100-pound graduations	0.094	0.020
Lettering "PPH x 1000"	0.094	0.016
Lettering "FUEL FLOW"	0.125	0.020
Shaded portion of pointer	------	------

3.5.2 Durable dull black. The marking "USAF" and "EFU-4/A" shall be permanently and legibly marked on the dial in 0.047 inch letters. This marking and all other markings, not otherwise specified, shall be finished in durable dull black.

3.6 Dial.

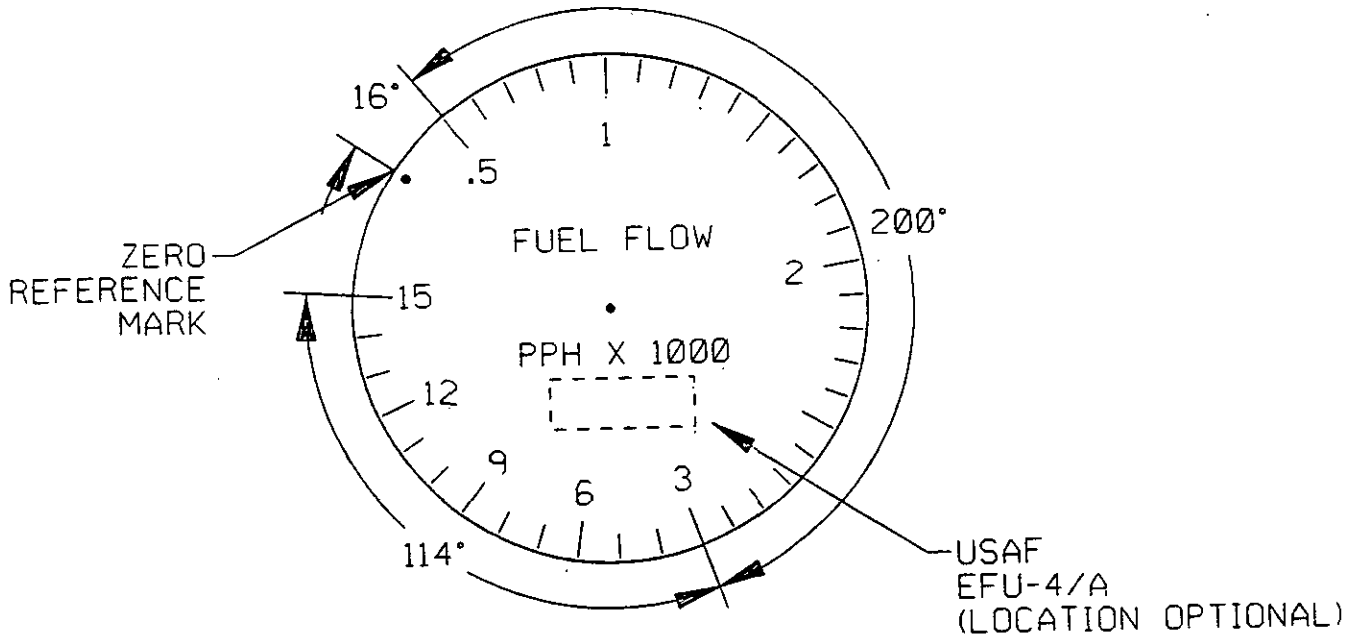
3.6.1 Dial diameter. The dial shall have a minimum diameter of 1.750 inches as measured at the outside ends of the scale graduations.

3.6.2 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be held to a minimum and shall not exceed 0.150 inch.

3.6.3 Electrical receptacle. The indicator electrical receptacle shall be in accordance with MS33678-18-1P, except it shall be designed for hermetic sealing.

3.6.4 Zero reference marking. The dial shall be provided with zero reference mark 16° before the first graduation of the dial. The zero reference mark shall be fluorescent-luminescent and shall be approximately 0.047 inch long, 0.016 inch wide, and shall be located on the outer edge of the dial as shown on figure 1 so that it is visible through the cover glass.

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TOLERANCE: ° +/- 15°

FIGURE 1. Indicator dial graduations

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3.7 Weight. The weight of the complete indicator shall not exceed 1.30 pounds.

3.8 Case. The case shall be hermetically sealed and shall conform to MS33639. The dimension "L" shall be 5 ±.25 inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 First article tests. For the purpose of this specification, the qualification tests specified in MIL-I-7566 shall be identified as first article tests.

4.2 General. The sampling, inspection, and test procedures shall be in accordance with MIL-I-7566.

4.3 Performance. The performance of the indicator shall be checked against tables I and II of this specification and shall not exceed the tolerances specified herein.

4.4 First article testing.

4.4.1 First article test sample tested by the contractor. The contractor shall subject three indicators to the first article test specified in 4.1.2.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 in accordance with MIL-P-116.

5.1.2 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved in accordance with MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

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TABLE I. Indicator fuel flow rate tolerances

FLOW RATE PPH	SCALE ERROR		TOLERANCE SUMMATION OUTPUT	PPH ± ERRORS**
	ROOM & HIGH TEMPERATURE	LOW TEMPERATURE	ROOM TEMPERATURE	HIGH & LOW TEMPERATURE
500	10	15	45	70
1,000	10	15	45	70
1,500	10	15	45	70
2,000	10	15	45	70
2,500	10	15	45	70
3,000	-10 +100	-15 +150	-45 +360	-70 +550
4,000	100	150	360	550
6,000	100	150	360	550
9,000	100	150	360	550
12,000	100	150	360	550
15,000	100	150	360	550

**The summation output error is the difference in pph between the actual potentiometer output signal and a master transmitter of negligible error.

TABLE II. Indicator test tolerances.

TEST	SCALE 500-3000 RANGE	ERROR 3000-15000 RANGE	TOLERANCE SUMMATION 500-3000 RANGE	PPH ± OUTPUT ERROR 3000-15000 RANGE
Frequency and Voltage variation	10	100	10	100
Position Error at 2,500 PPH	10	---	10	---
Friction:				
Room and High Temperature	10	100	10	100
Low Temperature	20	200	20	200
Vibration:				
Pointer Oscillation	20	200	20	200
Pointer Variation	20	200	20	200
Endurance:				
Response Time Three Second Maximum and Power Input 6.5 Watts Maximum	15	150	70	550

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5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A," "B," or "C" as specified (see 6.2).

5.2.1 Container-Cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit container shall be specified by the contracting activity. Unit container shall be large enough to allow for the application of cushioning/wrap material, of sufficient density and thickness, to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for the application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in "G's" established for the item.

5.2.2 Level A. Fiberboard containers do not meet level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size, and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedure and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked in accordance with special markings required by the contracting activity and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to ensure compliance with section 5 of this specification. Packaging tests shall be conducted in accordance with rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicator covered by this specification is intended for use to indicate the rate of, and to initiate signal for the summation of, flow of fuel in pounds per hour consumed by an aircraft turbo-jet engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODIIS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Levels of preservation, packaging, and packing required.

d. Number of first article samples and point of testing.

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6.3 Subject term (key words) listing. Recommended key words are:

EFU-4/A
Indicator
Rate of Flow
Totalizing
15,000 PPH

6.4 First article. When the first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.4.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Air Force - 99

Preparing activity:
Air Force - 71

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: MIL-I-25907C(USAF)	2. DOCUMENT DATE (YYMMDD): 94/06/30

3. DOCUMENT TITLE: Indicator, Rate Of Flow, EFU-4/A, Fuel

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME: OC-ALC/TICLA	b. TELEPHONE (Include Area Code) (1) Commercial: 405-736-5960 (2) AUTOVON: 336-5960
c. ADDRESS (Include Zip Code): 3001 Staff Drive, Tinker AFB, Ok 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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